أستاذة المادة: فاطمة أحمد

Waves and Ultrasound

WAVES

A wave is a disturbance which travels through a medium transferring energy from one point to another without causing any permanent displacement of the medium A wave motion is process of transferring a disturbance from one point to another without any transfer of particles of the medium.

Types of waves

Waves are broadly classified into two types

a. Based on the medium of propagation: mechanical wave and electromagnetic wave.

b. Based on the comparison of the wave direction with the direction of vibration of the particle: transverse wave and longitudinal wave.

Production and Propagation of waves: Based on the medium of propagation

1. **Production and propagation of mechanical waves**

A mechanical wave is the wave that requires material medium for its mode of propagation (or for it to transfer energy away from the source). Examples are waves travelling through springs, water waves, and sound waves

2. **Production and propagation of electromagnetic waves**

Electromagnetic waves are waves that do not need material medium for its mode of propagation (or for it to transfer energy away from the source). Examples are radio waves, visible light, ultra-violet rays, x-rays, gamma rays. Electromagnetic waves travel at the speed of light (3.0×108m). A wave which travels along a medium transferring energy from one part of the medium to another is called a progressive wave. The progressive wave can be divided into transverse and longitudinal waves

Production and Propagation of waves: Based on the comparison of the wave direction with the direction of vibration of the particle

1. Transverse waves

A transverse wave is a wave in which travel perpendicularly to the direction of the vibrations producing the waves.

2. Longitudinal wave

Longitudinal waves are waves which travel in a direction parallel to the vibrations of the medium

Waves Parameter

Wavelength, λ

Distance from a point on a wave to an identical point on the next wave. Measured in meters (m)

Amplitude

Size of maximum disturbance from the center line

Period, T

Time taken for one full wave to pass a point Measured in seconds (s)

Wave speed, v

Distance travelled by a wavelength in one second Measure in metres per second (m s-1)

$$\text{Wave speed} = \frac{\text{distancetravelled}}{\text{time}}$$

or

$$\text{Wave speed} = \frac{wavelength}{period}$$

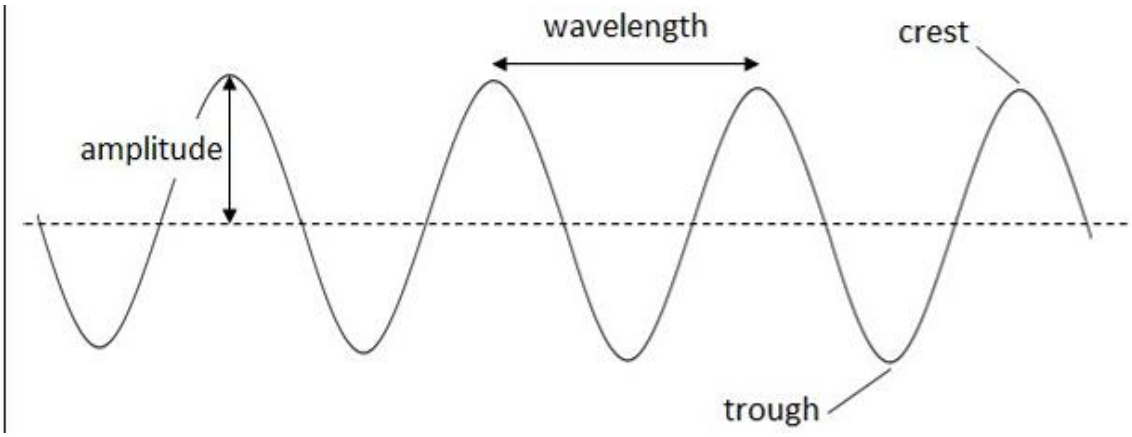
Frequency, f

Number of waves passing a fixed point every second Measured in hertz (Hz)

$$\text{Frequency} = \frac{\text{number of waves}}{\text{time}}$$

Or

$$\text{Frequency} = \frac{1}{\text{period}}$$



example

1. A radio station broadcasts at frequency of 300 KHz. If the speed of the wave is 3 x 108 ms-1 , calculate the period and wavelength of the wave?

$$f = 300\text{kHz} = 300000\text{Hz} = 3\times 10^{5}\text{Hz}$$

$$v = 3\times 10^{8} m/s$$

$$T = \frac{1}{f}$$

$$T = \frac{1}{3\times 10^{5}}$$

$$T = 3.3 \times 10^{-6}\ \text{S}$$

$$v = \frac{\lambda}{T}$$

$$\lambda = vT$$

$$\lambda = 3 \times 10^{8} \times 3.3 \times 10^{-6}$$

$$\lambda = 9.9 \times 10^{2}\ \text{m}$$

The Wave Equation

The wave equation links the speed, frequency and wavelength of a wave in one equation.

speed = frequency × wavelength

$$\boldsymbol{v = f\lambda}$$

example

An electromagnetic radiation has a speed of $3 \times 10^{8} m/s$ and a frequency of 106Hz, calculate its wavelength?

$$v = 3 \times 10^{8} \text{m/s}$$

$$f = 10^{6}\ \text{Hz}$$

$$v = f\lambda$$

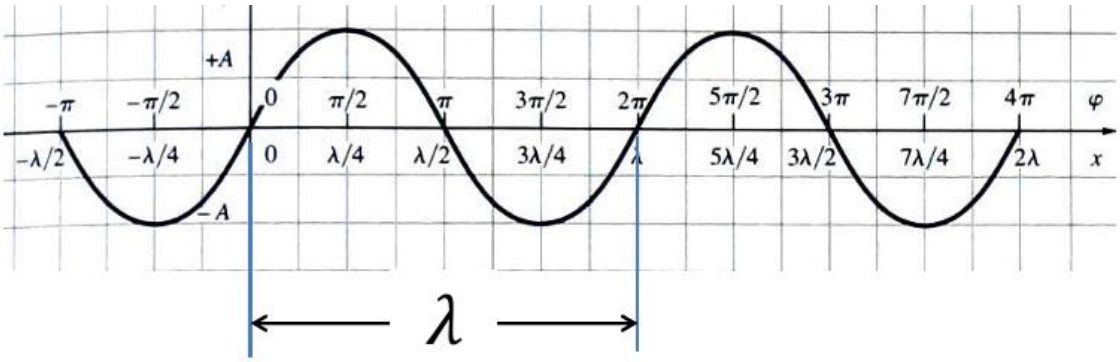
$$\lambda = \frac{v}{f}$$

$$\lambda = \frac{10^{8}}{10^{6}}$$

$$\lambda = 3 \times 10^{2}\ \text{m}$$

Harmonic Waves

Harmonic waves are those that have sinusoidal waveforms, i.e., can be described by sine or cosine**.**



One way to describe a harmonic wave**:**

$$y(x,t)=A\ sin\ (kx-\omega t+\varphi)$$

Where

A: Amplitude

φ:phase constant

K: Wave number is related to λ by

$$K=\frac{2\pi}{\lambda}$$

Where λ the Wavelength

ω: Angular frequency

$$\omega = 2\pi f$$

Where f the Frequency

Speed of harmonic wave

$$v = \frac{\omega}{k} = \frac{2\pi f}{\frac{2\pi}{\lambda}} = f\lambda$$

example

A radio waves transmitted from a certain radio station is represented by the wave equation: $y = 0.75\sin(0.67\pi x - 2 \times 10^8 \pi t)$ Calculate the (i) wavelength of the wave (ii) frequency of the wave (ii) velocity of the wave.

$$y = 0.75\sin(0.67\pi x - 2 \times 10^8 \pi t).....(1)$$

Calculate the wavelength λ

$$\boldsymbol{k}=\frac{\boldsymbol{2\pi}}{\boldsymbol{\lambda}}$$

$$\boldsymbol{\lambda}=\frac{\boldsymbol{2\pi}}{\boldsymbol{k}}$$

from equation (1)

$k=0.67\pi$

$\boldsymbol{\lambda}=\frac{\boldsymbol{2\,\pi}}{\boldsymbol{0.67\pi}}=2.96=3\text{m}$

Calculate the frequency *f*

$$\boldsymbol{\omega = 2\pi f}$$

$$\boldsymbol{f = \frac{\omega}{2\pi}}$$

From equation (1)

$\boldsymbol{\omega} = 10^8\,\boldsymbol{\pi}$

$$\boldsymbol{f} = \frac{10^8\boldsymbol{\pi}}{\boldsymbol{2\pi}} = 10^8\ \text{Hz}$$

Calculate the wave velocity *v*

$$v = f\lambda$$

$$v = (10^8)*(3) = 3\times 10^8\ \text{m/s}$$

The examples of Harmonic Waves :

1.Sound Waves (Pure Tones)

*Tuning Fork

*Musical Instruments (String vibrations)

2.Electromagnetic Waves

*Radio Waves

*Laser Light

3.Mechanical Waves

*Stretched Strings (Guitar or Violin strings)

*Springs (Oscillating mass)

4.Natural Phenomena

*Simple Water Ripples

*Clock Pendulum

Energy and Power in Wave Motion

Introduction

Wave motion is a fundamental concept in physics. Waves transfer **energy** from one place to another **without transporting matter**.

Examples:

- Sound waves in air
- Water waves in oceans
- Waves on a stretched string.

Energy in Wave Motion

When a wave travels through a medium, it carries energy. This energy is shared between:

a) Kinetic Energy (KE)

- Due to motion of particles
- Particles move up and down (oscillate)

b) Potential Energy (PE)

- Due to restoring forces (elastic forces)
- Example: stretched string or compressed medium

Total Energy:

$$E = KE + PE$$

---

3. Energy Dependence on Amplitude

One of the most important results:

$$E \propto A^2$$

Where:

- (A) = amplitude

This means:

- If amplitude doubles → energy becomes **4 times larger**

---

4. Energy in Waves on a String

For a wave traveling on a stretched string:

$$E \propto A^2 \omega^2$$

Where:

- ω = angular frequency

So energy depends on:

- Amplitude
- Frequency

---

5. Power of a Wave

Definition:

Power is the rate of energy transfer.

$$P = \frac{E}{t}$$

Where:

- (P) :power
- (E) : energy
- (t) : time

Power in a String Wave

$$P = \frac{1}{2}\mu\omega^2 A^2 v$$

Where:

- (μ): mass per unit length
- (ω): angular frequency
- (A) : amplitude
- (v) : wave speed

Important:

- Power increases with amplitude
- Power increases with frequency

Applications of Wave Energy

Sound Waves

- Carry energy through air
- Used in communication

Water Waves

- Ocean energy (wave power generation)

Light Waves

- Carry electromagnetic energy
- Used in solar power
